

Thrishul Reddy Poreddy | DATA ENGINEER

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PROFESSIONAL SUMMARY:

Data Engineer with 3+ years of experience designing, developing, and deploying robust and scalable data pipelines and solutions. Proven expertise in leveraging big data technologies such as Apache Spark, Kafka, Snowflake, and AWS ecosystem to extract, transform, and load large volumes of data from various sources. Strong analytical and problem-solving skills focused on data governance, security, and performance optimization. Adept at building and maintaining data warehouses, data lakes, and data marts to support business intelligence and advanced analytics. Experience in Agile methodologies and CI/CD pipelines for efficient and reliable data delivery.

TECHNICAL SKILLS

Data Processing: Apache Spark, Apache NiFi, Spark Streaming, Apache Airflow, Kafka

Data Warehousing: Snowflake, Amazon Redshift, Azure Synapse Analytics, Google BigQuery

Programming & Scripting: -Advanced Python libraries: Pandas, NumPy, PyArrow, Shell scripting for automation, SQL, Scala

Visualization: - Data storytelling through advanced Tableau/Power BI features, Superset

Big Data Tools - Apache Hive, HBase, Flink, Presto, Drill

ETL Tools: - dbt (Data Build Tool), Talend, Informatica, Apache Airflow, Apache NiFi, AWS Step Functions

Streaming and Real-Time Processing: - AWS Kinesis, Azure Event Hubs, Confluent Kafka

Performance Tuning: SQL Query Optimization, Hadoop Optimization, Spark Job Tuning

Cloud Technologies: AWS: - S3, Glue, EMR, Athena, Kinesis, Lambda
Azure: - Data Factory, Databricks, Key Vault, Data Lake Storage

Data Governance: SOX Compliance, GDPR Compliance, Data Protection Regulations

Data Modeling & Storage: - Parquet, Avro, ORC (optimized data formats), Logical and physical data modeling tools like ER/Studio, Erwin, Relational and NoSQL databases (PostgreSQL, MongoDB)

Machine Learning: Scikit-learn, Predictive Modeling

CI/CD Tools: Jenkins, Docker, Kubernetes

Monitoring & Logging: Splunk, ELK Stack, CloudWatch, Azure Monitor, Prometheus, Grafana

Security & Governance: IAM (Identity and Access Management), Encryption techniques (AES, TLS), Key Management Systems (e.g., AWS KMS, Azure Key Vault)

Soft Skills: Agile Methodology (Scrum frameworks), JIRA, Trello, Cross-functional team collaboration

Testing: - Unit testing for data pipelines (Great Expectations)

Data Integration: RESTful APIs, Third-party Platform Integration

Financial Analysis: Bloomberg Terminal, QuantLib

CI/CD & Automation: - Git, GitHub Actions, Terraform

PROFESSIONAL EXPERIENCE:

Charles Schwab – TX | Data Engineer | Jan 2024 – Current

- Leveraged Apache Spark to handle distributed data processing, achieving a 40% improvement in performance for complex analytics queries. Built and optimized ETL pipelines using Apache Airflow to process financial data daily, reducing data processing time by 30%.
- Designed and maintained a Snowflake-based data warehouse integrated with Azure Synapse Analytics, improving data query efficiency by 25%. Performed advanced analytics using Google BigQuery and Azure Synapse Analytics for strategic decision-making.
- Created interactive dashboards in Tableau, enabling real-time monitoring of Key Performance Indicators (KPIs) and reducing reporting errors by 15%.
- Worked on data governance initiatives ensuring compliance with SOX and GDPR within the Indian American Banking sector, minimizing regulatory risks & fines. Implemented high-performance data processing pipelines leveraging Apache Spark to handle terabytes of financial data efficiently.
- Streamlined deployment processes by implementing CI/CD pipelines using Jenkins, Docker and Kubernetes, achieving a 50% reduction in deployment time and increasing release frequency. Deployed scalable data pipelines using Python, SQL & Scala for data ingestion, transformation and loading (ETL).
- Conducted in-depth financial impact analysis utilizing Bloomberg Terminal and QuantLib to evaluate the economic viability of new products, optimize trading strategies, and assess risk exposures. Deployed predictive credit risk models using Scikit-learn, resulting in a 20% reduction in default rates.
- Enhanced system reliability and pro-activeness by developing and implementing real-time monitoring and anomaly detection solutions using Splunk and the ELK Stack, enabling rapid identification and resolution of critical issues.

Wipro – India | Data Engineer | Oct 2020 – Dec 2022

- Achieved a 40% reduction in ETL processing time by strategically implementing Apache NiFi and Spark Streaming, enhancing data pipeline efficiency for real-time analytics.
- Engineered robust real-time data processing frameworks utilizing Apache Kafka and Spark Streaming, enabling immediate insights and decision-making capabilities for retail operations.
- Implemented star & snowflake schemas in Amazon Redshift, facilitating optimized data retrieval and analytical performance for complex retail queries.
- Conducted in-depth performance tuning of SQL queries, resulting in a 35% decrease in query execution time, thereby improving the overall responsiveness of data-driven applications.
- Seamlessly integrated diverse data sources, including RESTful APIs and third-party platforms, to create a unified data ecosystem that supports comprehensive analytics and reporting.
- Automated complex data workflows using Apache Airflow and AWS Step Functions, significantly reducing manual intervention and enhancing operational efficiency across data processing tasks.
- Executed performance tuning for Hadoop and Spark jobs, leading to substantial improvements in data processing speeds and resource utilization, optimizing the data engineering infrastructure.
- Developed interactive dashboards using Power BI to visualize key retail metrics, empowering stakeholders with actionable insights and facilitating data-driven decision-making.
- Ensured adherence to data protection regulations, including GDPR and local data privacy laws, by implementing best practices in data governance and security protocols.
- Actively participated in agile project management practices using tools like JIRA and Trello, contributing to sprint planning, backlog grooming and cross-functional collaboration to deliver high-quality data solutions.

EDUCATION and CERTIFICATIONS:

Master of Computer Science - Northern Arizona University, Flagstaff, AZ, USA.

Bachelor of Electronics and Communication Engineering – Jawaharlal Nehru Technological University, Hyderabad, India.

Certified AWS Data Engineer Associate – Amazon | **Certified Azure Data Engineer Associate** – Microsoft